

Candidate Intelligence Scoring System

Design Principles, Signal Architecture & Scoring Logic

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1. Core Design Philosophy

This system exists to surface observable facts and behaviors about candidates — not to score brief performances or validate self-reported claims. Every signal in the model is tied to a specific moment in the process and a specific observable action.

The system separates three things that most hiring processes collapse together:

- What the candidate is actually showing us (Candidate Intelligence)
- Whether our process is running efficiently (Workflow Intelligence)
- Whether our decisions are consistent with the evidence (Decision Quality)

Scoring = observed facts and behaviors only. Interviewer opinion is intentionally excluded from construct scores so that the four construct scores remain auditable, comparable, and free from human judgment contamination. Interviewer impression feeds the Decision Quality dashboard exclusively, where it can be analyzed for drift, bias, and correlation with behavioral signals.

2. The Four Constructs

The scoring model is grounded in I/O psychology research showing these four dimensions reliably predict job performance. Each construct is independently scored and then weighted into an overall candidate score.

Construct	Weight	Scientific Basis
Capability	40%	Strongest predictor of job performance (Schmidt & Hunter, 1998)
Conscientiousness	25%	Consistent predictor across roles and industries (Barrick & Mount, 1991)
Communication	15%	Predicts teamwork, stakeholder management, leadership (Riggio & Feldman, 2005)
Motivation	20%	Self-selection and engagement predict discretionary effort and retention — floor of 0.2 applied if application_submitted event exists in window

Note on Communication: this construct was originally labeled Interpersonal but renamed for readability. The scientific grounding remains Riggio & Feldman (2005) on interpersonal effectiveness predicting teamwork, stakeholder management, and leadership.

3. Written vs. Verbal Communication — A Critical Distinction

Communication signals split into two fundamentally different dimensions that score into different constructs:

- Written quality (typos, grammar, syntax, clarity, formatting) scores primarily into Capability — because written precision signals cognitive precision, not interpersonal skill.
- Verbal quality (interview communication, tone, fluency, question quality) scores into Communication — because it reflects interpersonal effectiveness with real humans in real time.
- Evidence that demonstrates both (authored articles, public speaking, video submissions) can score into both constructs on different dimensions — content scores Capability, delivery quality scores Communication.

The same piece of evidence evaluated on two distinct dimensions is not double-counting. Content quality is a different signal from communication effectiveness. Each rolls up to its own construct stack independently.

4. Observable Evidence Over Self-Reported Claims

AI-generated resumes have degraded the signal value of keyword matching, self-reported skills, and resume content generally. This system is designed around that reality.

4.1 Employer Research

When an application is submitted, the Screening A[i]gent researches each past employer directly rather than taking resume claims at face value:

- Industry and company size/environment match
- Actual tech stack the employer used — validated against what the candidate claims
- Role similarity derived from researching actual job descriptions at prior employers, not just comparing titles (titles vary dramatically across organizations)
- For sales roles: audience, product type, sales cycle length, quota and volume comparability
- Tenure: time spent per employer, surfaced as context with configurable job-hopping thresholds per hiring manager preference — not a baked-in score penalty

4.2 Tech Stack Validation

If claimed tech experience cannot be verified against employer adoption timelines, the system does not penalize the candidate's score. Instead it generates a query flag in the recruiter widget with a suggested clarification question for the screening interview. The interview is the right place to resolve ambiguity — not the scoring layer.

4.3 Public Evidence

The agent researches publicly available evidence including community forum mentions, public speaking engagements, authored articles and research, and portfolio work. Important scoring rules:

- Positive public evidence (organic forum mentions, speaking engagements) adds to score — absence does not reduce it. Most strong candidates are simply not publicly visible.

- LinkedIn recommendations are validated for trading patterns. If trading is detected, the signal is silently nulled — no score impact in either direction, not surfaced in the widget.
- LinkedIn profile sentiment is treated as context only and excluded from scoring — profiles are heavily optimized and often ghostwritten, making signal quality unreliable.

5. The Self-Selection Principle

One of the most predictive and least measured signals in hiring is voluntary, uncoerced behavior — what a candidate chooses to do when nothing is required of them.

"At a deeplearning.ai AI in TA event deliberately scheduled on a Friday evening, only candidates who are genuinely passionate about the subject show up. The selection happens automatically. No test required." The opt-in assignment applies the same logic: give candidates room to demonstrate who they are on their own terms, then observe what they do with that room.

Why opt-in behavior is high signal:

- Uncoerced — no external pressure to perform
- Self-directed — chosen without being asked
- Consistent with real work conditions — top performers self-assign stretch work
- Observable and factual — did they or did they not

A single opt-in event (e.g., submitting an optional task assignment) scores across three constructs on distinct dimensions:

Signal / Category	Primary Construct	Notes
Choosing to opt in	Motivation	<i>Boolean — did/did not. Not additive across multiple opt-ins.</i>
Quality of the submission	Capability	<i>What they produced demonstrates cognitive precision and skill</i>
Care and thoroughness in submission	Conscientiousness	<i>How they did it — attention to detail, completeness, follow-through</i>

Opt-in scoring is boolean per type — the signal is the choice to engage, not the volume of engagements. Stacking multiple opt-ins does not inflate the Motivation score.

6. Stage 1 Signal Map — application_submitted

Trigger: ATS webhook fires Apps Script, which deploys the Screening A[i]gent.

Signal / Category	Primary Construct	Notes
Employer industry & size match	Capability	<i>Agent-researched, not self-reported</i>

Role similarity (JD-level research)	Capability	<i>Titles don't transfer — functions do</i>
Tech stack validation	Capability	<i>Flags query if unverifiable; does not score negatively</i>
Years of experience (verified)	Capability	<i>Validated against employer timeline</i>
Portfolio / public evidence found	Capability	<i>Agent-sourced; absence neutral</i>
Authored articles / research	Capability + Communication	<i>Content = Capability; writing quality = Communication</i>
Public speaking engagements	Communication	<i>Verbal effectiveness evidence</i>
Community forum mentions (organic)	Capability	<i>Additive only; absence neutral</i>
Application completeness (required fields)	Conscientiousness	<i>Did candidate follow instructions on required items</i>
Completeness beyond required fields	Conscientiousness + Motivation	<i>Going beyond minimum is a dual signal</i>
Opt-in submissions (video, AI interview, assessment)	Motivation + Capability + Conscientiousness	<i>Choice = Motivation; quality = Capability; care = Conscientiousness</i>
Time spent on site / application	Motivation	<i>Investment of time before submitting</i>
Portal / materials engagement	Motivation	<i>Research behavior before applying</i>
Certifications & education	Capability	<i>Verified credentials only</i>
Written application quality (if scored)	Capability	<i>Cognitive precision in written communication</i>
Video intro / AI interview (if submitted)	Communication	<i>Verbal and interpersonal quality</i>

Agent output to recruiter widget: construct scores for signals observable at this stage, Advance / HM Review / Do Not Advance recommendation with plain-language reasoning, ask-next prompts for flagged items. Any recruiter override requires a recorded reason which feeds Decision Quality only.

7. Stage 2 — Post-Screening Disposition & Candidate Signals

After the Screening A[i]gent returns its recommendation, the recruiter makes a disposition decision in the ATS widget. The agent produces one of three recommendations: Advance, HM Review, or Pass.

7.1 The Three Recommendation Outputs

Advance — recruiter should move candidate to recruiter interview. HM Review — an additional decision support layer where the hiring manager reviews the candidate and returns a recommendation to the recruiter before any interview decision is made. The recruiter still owns the final disposition. Pass — do not advance.

7.2 Disposition Logic & Override Rules

Any time a recruiter acts against the agent recommendation, they must record a reason before the action is available. The same applies when a hiring manager's recommendation diverges from the agent

recommendation. All of this is logged to ATS and data store and feeds Decision Quality only — it does not affect candidate construct scores.

Signal / Category	Primary Construct	Notes
Agent rec: Advance → Recruiter advances	Decision Quality	<i>Aligned — no override note required</i>
Agent rec: Advance → Recruiter passes	Decision Quality	<i>Override — recruiter must record reason</i>
Agent rec: Advance → Recruiter sends to HM Review	Decision Quality	<i>Override — recruiter must record reason</i>
Agent rec: HM Review → Recruiter forwards to HM	Decision Quality	<i>Aligned — no override note required</i>
Agent rec: HM Review → Recruiter advances directly	Decision Quality	<i>Override — recruiter must record reason</i>
Agent rec: HM Review → Recruiter passes	Decision Quality	<i>Override — recruiter must record reason</i>
Agent rec: Pass → Recruiter passes	Decision Quality	<i>Aligned — no override note required</i>
Agent rec: Pass → Recruiter advances	Decision Quality	<i>Override — recruiter must record reason</i>
Agent rec: Pass → Recruiter sends to HM Review	Decision Quality	<i>Override — recruiter must record reason</i>
HM recommendation diverges from agent rec	Decision Quality	<i>HM must record reason — logged separately</i>

7.3 HM Review — Decision Support Layer

HM Review is not a routing shortcut that bypasses the recruiter screen. It is an additional decision support layer. The flow is: agent recommends HM Review → recruiter forwards to HM → HM reviews and returns their own recommendation → if HM diverges from agent recommendation, HM records reason → recommendation returns to recruiter → recruiter makes final disposition, recording reason if they diverge from agent recommendation. The recruiter always owns the final call.

HM Review adds informed input, not a bypass. The recruiter remains the decision owner at every stage.

7.4 Pass Disposition — Rejection Timing

When a candidate is passed, the rejection email is queued with a minimum 7-day delay from last interaction before sending. This is a candidate experience and employer brand design decision — instant automated rejections signal that the candidate was not genuinely considered, which carries legal and reputational risk. Seven days is fast by market standards but maintains dignity. The delay is configurable per org protocol; the minimum floor is not.

7.5 Workflow Time-in-Stage Clock

The workflow clock starts at application_submitted and stops at the first disposition decision. The next stage clock starts at that decision and stops at the next. This creates a continuous chain of stage durations feeding Workflow Intelligence — not candidate scoring.

7.6 Keep-Warm Communications & Candidate Signals

Candidates who are waiting without a decision receive automated keep-warm communications. Engagement with these communications generates candidate signals that feed Motivation scoring.

Signal / Category	Primary Construct	Notes
interview_invited timestamp	Conscientiousness	<i>Starts responsiveness clock</i>
interview_rsvp timestamp	Conscientiousness	<i>Stops clock — delta in hours = responsiveness signal</i>
Keep-warm email open / click	Motivation	<i>Engagement under zero pressure — self-selection signal</i>
Accesses linked resource, article, chatbot	Motivation	<i>Opt-in research behavior while waiting</i>
Joins webinar or free trial from keep-warm	Motivation	<i>High-investment opt-in — strong self-selection signal</i>
Completes optional assessment during wait	Motivation + Capability + Conscientiousness	<i>Choice = Motivation; quality = Capability; care = Conscientiousness</i>

Keep-warm trigger rules: candidates in application stage with no decision after 14 days receive a keep-warm touchpoint with org-configurable content — prep resources, company information, webinars, free trials, interview chatbot, feedback survey. Candidates who have completed at least one interview with no decision receive a keep-warm every 7 days until decision is recorded. After week 3 with no decision, keep-warm includes contact information for the person responsible for the delay. This creates accountability without requiring the system to enforce it directly.

The keep-warm architecture serves three purposes simultaneously: candidate experience (they feel valued and informed), employer brand (the org is perceived as respectful and communicative), and signal collection (engagement with keep-warm content is genuine motivation data generated under no pressure).

8. Flag Resolution Architecture

When the Screening A[i]gent identifies a signal that requires human clarification (e.g., unverifiable tech stack claim), rather than storing a passive flag, the system triggers an automated follow-up sequence:

- Flag triggers automated email to candidate requesting clarification on the specific item
- Same email offers optional task assignment or assessment submission
- Candidate response resolves the flag directly in the data model via a consistency field

- If response clears the flag: recruiter widget updates with refined recommendation
- If response confirms a disqualifying gap: Do Not Advance without consuming recruiter interview bandwidth

This closed loop is the difference between a flag that informs a decision and a flag that just sits in a field. The consistency field stores the flag, the response, and the resolution status — creating an auditable trail that feeds Decision Quality.

8. Interviewer Score Isolation & Scorecard Visibility

This section covers two related governance decisions that protect the integrity of evaluation data. Both are configurable — the system is designed to flex to an organization's existing tech stack and culture. But the science is documented here so that design choices are made with full information, not just organizational convenience.

8.1 Interviewer Scores Are Excluded From Construct Scoring Entirely

Interviewer impression scores do not feed any of the four construct scores — Capability, Conscientiousness, Communication, or Motivation. This is an intentional architectural decision, not a limitation.

Reasons:

- Construct scores are built on observable, factual behavioral signals. Interviewer impression is opinion — even well-intentioned, structured opinion. Mixing opinion into a factual scoring layer makes the scores unauditable.
- Excluding interviewer scores preserves the ability to audit decisions against evidence. If an interviewer rated a candidate poorly but all behavioral signals were strong, that discrepancy is visible and investigable. If interviewer opinion contaminated the construct scores, the discrepancy disappears.
- It enables genuine bias detection. You cannot measure whether interviewers are introducing bias if their judgments are already baked into the score you are comparing their judgments against.

Where interviewer scores do appear:

- Decision Quality dashboard — for leadership and TA Ops to analyze interviewer scoring patterns, consistency, drift, and correlation with behavioral signals over time
- Executive reporting — aggregate interviewer reliability metrics
- Calibration workflows — to identify interviewers whose scores consistently diverge from signal evidence

The behavioral signal score tells you what the candidate showed. The interviewer score tells you what the interviewer thought. Keeping those two things separate is what makes the audit possible.

8.2 Interviewers Do Not See Each Other's Scores Before Submitting

Interviewers submit scorecards independently, without visibility into other interviewers' scores or ratings before their own submission is complete. This applies across all interview stages.

The science behind this:

- Anchoring bias — the first score submitted becomes a psychological reference point for all subsequent evaluators, even when reviewers believe they are forming independent judgments (Tversky & Kahneman, 1974)
- Cascade effect — in panel settings, early positive or negative impressions propagate through the group, compressing score variance and reducing the diagnostic value of having multiple evaluators
- Independent evaluation is the entire point of a panel — if evaluators can see each other's scores first, you are effectively running a single evaluation with social pressure attached

8.3 Recruiter Scorecard Visibility to Hiring Managers — Optional Design

Whether hiring managers should see recruiter scorecards before conducting and submitting their own interview scorecard is an org-specific decision. This system supports either configuration. The tradeoffs are documented here for informed design.

The case for isolation (default recommendation):

- The HM's independent read is more valuable than a pre-anchored read — for the same anchoring bias reasons as Section 8.2
- If the recruiter's assessment is visible before the HM interview, the HM walks in with a frame that shapes what they notice and how they interpret what they hear
- The right pre-interview context for an HM is behavioral signal data — what the system observed factually about the candidate — not the recruiter's opinion. This gives preparation without anchoring judgment.

The case for sharing (common org preference):

- HMs often want context before investing interview time, particularly for senior or complex roles
- In orgs with strong recruiter-HM trust and calibration, the recruiter assessment can be genuinely useful preparation
- Practically, in many organizations this will not be accepted as a process change regardless of the science

The system is flexible and built around what works for each organization. But the science is here if you want to fight with data. Independent evaluation produces more reliable signal. Anchored evaluation produces faster consensus — which is not the same thing.

Configurable options this system supports:

- Full isolation: no scorecard visibility until all scorecards for that stage are submitted
- Partial isolation: recruiter scorecard locked from HM view until HM scorecard submitted, then both released simultaneously
- Open: all scorecards visible in real time (not recommended but supported)
- Role-based: isolation applied for senior/executive roles only, open for high-volume roles

9. Assumed Infrastructure — Data Privacy & Informed Consent

This system assumes that data privacy disclosure and informed consent are handled at application submission as baseline infrastructure. This is not optional and is not documented here in detail because it should already exist in any compliant hiring operation before an AI-assisted scoring system is layered on top.

At minimum, every application submission should include:

- Disclosure that AI is used to assist in screening and evaluation
- Plain-language description of what data is collected and how it is used
- Candidate rights to access, correct, and request deletion of their personal data
- How to exercise those rights and who to contact
- Informed consent to proceed — candidates choose to continue with full knowledge

No one reads it. Put it there anyway. The audit trail is for you, not them.

9.1 Regulatory Context (Not Legal Advice)

The following frameworks are relevant depending on where candidates are located and where the hiring organization operates. Consult counsel for implementation specifics.

- **Article 22 prohibits solely automated decisions with significant effects on individuals. Human-in-the-loop with override authority satisfies this. Candidates have the right to request access to their data and explanation of how automated processing affected decisions about them. Data retention limits apply — candidate records cannot be stored indefinitely without a legal basis.**GDPR (EU/EEA) —
- **Finalized ADMT regulations take effect January 1, 2027. Requires pre-use notice of AI in employment decisions, risk assessments, and honoring candidate access rights. Human review with documented authority to override satisfies the meaningful human involvement standard.**CCPA / CPRA (California) —
- **AI hiring tools classified as high-risk as of August 2024. Requires transparency, human oversight, bias mitigation, and technical documentation. Applies to non-EU companies hiring EU residents.**EU AI Act —
- **Colorado, Illinois, New York City, and a growing number of states have enacted or are enacting AI hiring regulations. The trend is consistently toward more disclosure, more human oversight, and more candidate rights.**State-level US laws —

This system's architecture — human confirmation required for every disposition, mandatory override notes, full audit trail in ATS and data store, AI recommendation as decision support not final decision — is designed to satisfy human-in-the-loop requirements across all of these frameworks. Orgs implementing this system should conduct a Data Protection Impact Assessment (DPIA) appropriate to their jurisdiction before deployment.

10. Agent Architecture — AI-Led Research With Human-In-The-Loop Decisions

The three agents in this system are browser extensions that operate autonomously and write their results to the ATS. This is a critical design distinction that defines the system's philosophy.

This is not AI-assisted recruiting. It is AI-led research with human-in-the-loop decisions. Assisted implies the human is still doing the work with help. Led means the AI completes the research autonomously and returns a result for human review and decision. The recruiter is the decision-maker, not the researcher.

The ATS widget is the display and decision layer. The browser extensions are the intelligence layer that populates it. Because the intelligence lives in the extensions rather than inside the ATS, the system is ATS-agnostic — swap Greenhouse for any other ATS and the agents continue to function.

Agent	When It Runs	What It Does Autonomously
Screening A[i]gent	application_submitted	Researches candidate across LinkedIn, employer sites, public forums, GitHub, speaking events, published articles, community mentions. Returns complete intelligence package to ATS widget: verified experience, role similarity, tech stack validation, public evidence, query flags, ask-next prompts, recommendation. Recruiter never opens a tab.
Interview A[i]gent	During every interview stage	Listens and transcribes via AI notetaker. Scores against rubric in real time. Surfaces follow-up question suggestions in ATS widget. Writes scorecard to candidate file automatically post-interview. Recalculates construct scores and generates next-steps recommendation. Recruiter focuses entirely on the candidate.
Offer A[i]gent	Pre-offer through DocuSign	Surfaces prior signal intel and compensation data before the offer call. Flags timing issues (e.g. commission forfeiture), competing processes, location and visa requirements. Auto-populates agreed terms during the call. Routes approvals. Triggers DocuSign and onboarding on completion. Recruiter walks into the offer call knowing what this specific candidate needs to say yes.
Sourcing A[i]gent	Open req / search trigger	Researches and surfaces candidates autonomously. Returns shortlist to recruiter with intelligence already attached. No manual research required.
Listening A[i]gent	Post-hire ongoing	Monitors signals across the work environment passively — meeting transcripts, communication patterns, sentiment trajectory, culture indicators, discretionary effort, learning engagement. Does not ask how people feel. Observes what they do. Early attrition signal before the conscious decision to leave. Feeds Employee Intelligence and Decision Quality feedback loop.

11. What This System Is Not

- Not keyword matching — AI resumes have degraded that signal
- Not resume scoring — resumes are a starting point for investigation, not a source of truth

- Not brief performance evaluation — the system scores patterns across the full process, not single moments
- Not opinion-based — interviewer impression is deliberately excluded from construct scores
- Not punitive for absence of public visibility — only positive public evidence adds to score

This document is a living design spec. Stages 2+ to be added as mapping continues.